Patrick Liu

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Education

University of California, Santa Barbara GPA: 4.00
B.S. in Computer Engineering *Expected Graduation: June 2027*
Relevant Coursework: Data Structures and Algorithms, Machine Learning, Computer Networks, Artificial Intelligence, Computer Architecture, Object-Oriented Programming, Discrete Mathematics, Statistics, Linear Algebra
Awards: Promise Scholar, Engineering Honors Program, Dean's List, ACM Most Innovative Project

Experience

Network Consultant December 2025 – Present
UC Santa Barbara Information Technology *Santa Barbara, CA*
- Troubleshot wired and wireless connectivity issues across campus residential network infrastructure of 10K+ users
- Diagnosed DNS, DHCP, and TCP/IP routing issues by debugging network configurations and hardware

Software Engineering Intern June 2025 – September 2025
Lookout *Boston, MA*
- Cut infrastructure costs 30% and data access time 20% by migrating from Drive to Firebase RTDB and Storage
- Owned edge-to-cloud video pipeline on GCP VMs with systemd, achieving 99% uptime across 100+ devices
- Created a journey replay platform with React, Firebase Auth, RTDB, and Cloud Storage, saving 100+ hours
- Built CI/CD pipelines with GitHub Actions, reducing deployment time from 2 hours to 10 minutes

Software Engineering Intern June 2024 – August 2024
Lookout *Boston, MA*
- Built RESTful CRUD APIs on Firebase Realtime Database and Storage using object-oriented design
- Expanded object detection coverage by labeling new classes and retraining YOLO models with PyTorch
- Containerized a Python media service with Docker, cutting deployment setup time by 60% across 100+ devices

Software Engineering Intern June 2023 – August 2023
Laminar *Somerville, MA*
- Built a Python microservice on AWS EC2, IAM, and S3 to process sensor data for machine learning pipelines
- Reduced regressions 20% by adding Pytest tests for core APIs, targeting gaps found via failure logs
- Collaborated with teams to define requirements, prioritize fixes, and deliver customer-facing improvements

Projects

Cognix (SB Hacks Best in Education — 1st of 55 Projects) | *Next.js, TypeScript, MongoDB*
- Created a VS Code extension with LLM guardrails to enforce course AI policies during coding workflows
- Built Next.js and MongoDB APIs to process IDE telemetry and surface instructor-facing session insights
- Designed real-time LLM pipelines to analyze IDE sessions, score originality, and flag policy violations

Cadence (MakeMIT x Harvard 2nd Overall, Best Use of Vultr) | *PostgreSQL, Node.js, Express.js*
- Built a real-time platform with Node.js and Next.js to stream and visualize live instrument sensor data
- Designed ingestion pipelines with Express and Serial I/O, streaming 1K+ events/session into PostgreSQL
- Exposed REST and WebSocket APIs for low-latency streaming and real-time performance visualization

Bentley Library | *Python, Django, Postgres, Go, Auth0, GPT-OSS*
- Created a Django library platform for 400+ users with catalog search, circulation, holds, and account workflows
- Reduced search latency 60% by replacing ILIKE scans with GIN full-text search across 1K+ Postgres records
- Built a retrieval pipeline over 1K+ records using GPT-OSS query parsing and Go reranking for recommendations

Technical Skills

Languages: Python, C++, C#, SQL, Go, Java, C, JavaScript, TypeScript, HTML/CSS
Frameworks: React, Next.js, Node.js, Express.js, Flask, Django, Three.js
Tools: Git, Linux, Docker, Kubernetes, Firebase, MongoDB, PostgreSQL, GCP, AWS, Auth0, REST APIs, CI/CD